Centering the error makes the posterior mean optimal

Average the centered expansion Previous slide: pointwise squared-error expansion

$$\|z-a\|^2 = \|r\|^2 + 2r^T(m-a) + \|m-a\|^2$$

Take $E[\cdot\,|y]$; $m-a$ is fixed

$$E[r|y] = E[z|y] - m = 0$$

$$E\big[r^T(m-a)|y\big] = E[r|y]^T(m-a) = 0$$

$$V = \mathrm{Cov}[z|y]$$

$$E\big[\|r\|^2|y\big] = \sum_{j=1}^{d} E\big[r_j^2|y\big] = \mathrm{tr}\, V$$

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Conditional squared-error risk

$$R(a|y) = \mathrm{tr}\, V + \|m-a\|^2$$

- Risk is residual uncertainty plus squared distance from the posterior mean.

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Conditional squared-error risk

$$R(a|y) = \mathrm{tr}\, V + \|m-a\|^2$$
$$\|m-a\|^2 \geq 0 \quad \text{equality iff} \quad a = m$$
$$a^*(y) = m = E[z|y]$$
$$\min_a R(a|y) = \mathrm{tr}\, V$$

Earlier Gaussian

$$a^*(y) = \mu_{z|y}$$
$$\min_a R(a|y) = \mathrm{tr}\, \Sigma_{z|y}$$

- Risk is residual uncertainty plus squared distance from the posterior mean.
- The posterior mean minimizes model expected squared error; covariance measures what remains.